

Greta Ru-Mei Zu

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WORK EXPERIENCE

Artificial Intelligence Software Developer Intern

Montreal, QC, Canada

Ericsson Canada

Jan 2026 - May 2026

- Implemented and deployed **RAG** (retrieval augmented generation) chatbot using Hopsworks, DeepEval, and EricAI models.
- Enhancing a dimensioning tool (CANDI) with AI/ML capabilities and automating the building of the prediction model.
- Designed and deployed a price-predictor tool with regex and LLM process, and trained a LightBGM model for trend prediction.

Machine Learning Intern

Montreal, QC, Canada

Ericsson Canada

May 2025 - December 2025

- Designed and deployed **RAG** pipelines using **AWS Lambda**, **Bedrock**, and **S3**.
- Created Python-based agentic workflows to generate datasets for fine-tuning; improved workflow latency & results by ~40%.
- Designed and implemented a CLI based coding assistant using Langchain DeepAgents.

RESEARCH EXPERIENCE

Watermarking for Free-rider Detection in Federated Learning

Lausanne, Switzerland

Summer@EPFL - EPFL SaCS Lab with Professor Anne-Marie Kermarrec

June 2026 - August 2026

- Analysis and implementation of output layer watermarking schemes as a solution for free-rider detection in federated learning & design of a free-rider attack to prove the fragility of the output layer schemes. Experimental and theoretical proofs of the inefficiency of output layer watermarking for free-rider detection in our paper (SatML 2027).

Machine Learning algorithms & Energy efficiency

Montreal, QC, Canada

McGill DISC Lab with Professor Oana Balmau & Professor Bettina Kemme

June 2025 - December 2025

- Instrumented Switch-Transformers and Qwen **Mixture-of-Experts** models using CodeCarbon on DeepSpeed; profiled GPU energy consumption and identified optimization strategies that reduced energy usage by ~66% without loss in accuracy.
- Prepared a project for students for a Responsible AI course, including starter code and setup based on the research project.

TikTok algorithm analysis

Montreal, QC, Canada

The Prometheus Lab with Professor Joseph Vybihal

Sept 2024 - December 2024

- Designed Python-based bots to collect 400+ TikTok videos while bypassing scraping restrictions; conducted statistical analysis using Pandas and Scikit-learn to quantify algorithmic bias in engagement rates by user gender and content sentiment.

PROJECTS

Compiler - Compiler Design term project

Jan 2025 - May 2025

- Designed and created a full compiler from scratch for Mini-C language (subset of C) using Java and Bash scripts for automation.

SOC-cessful Schedule - Web Development term project

Nov 2024 - Jan 2025

- Designed and created a full-stack website for a booking tool used by school staff and students (*XAMPP stack, SQLite3 database*).

Paxos Total Order Game - Distributed Systems project

Nov 2024 - Dec 2024

- Using the Paxos Consensus Algorithm to implement total order for a Java-based multiplayer distributed systems game.

Travel Agency Management System - Database Systems project

Jan 2024 - May 2024

- Developed a Java-based application for a travelling agency, to book flights, hotels or rent cars. Created relational schemas, E/R diagrams, *SQL* queries and *JDBC (Java)* to provide a *database* and functions to navigate and use it.

Artificial Intelligence Agent - AI course term project

Oct 2023 - Dec 2023

- Developed a Python-based agent to play against another player in a game using techniques such as *Monte Carlo Tree Search*, *A* Search* and *heuristics* to efficiently make decisions and win against any opponent; winning top 20% in the class tournament.

McWics Hackathon project - BCV

2025

- Designed a full-stack web-app (*React, Drizzle, Postgresql*) to improve CV with most relevant experience using Gemini API calls.

EDUCATION

Imperial College London

London, UK

- Postgraduate taught MSc in Computing (Software Engineering)

Sept 2026 - Sept 2027

McGill University

Montreal, QC, Canada

- Bachelor of Science in Computer Science (Artificial Intelligence)

August 2022 - May 2026

- *Alma Mater Scholarship*, awarded by McGill University in 2022

GPA: 3.85/4.0

- Relevant Coursework: Algorithms & Data structures, Artificial Intelligence, Applied Machine Learning, Reinforcement Learning, Software Systems, Operating Systems, Compiler Design, Database Systems, Data Science, Distributed Systems

SKILLS & INTERESTS

Programming Languages: Java, Python, C/C++, Bash, Assembly, SQL, Javascript, PHP, OCaml, WebDev languages (HTML/CSS)

Frameworks & Libraries: AWS, Docker, Git, Pandas, NumPy, Matplotlib, Pytorch, TensorFlow, Scikit-learn, CodeCarbon, SQLite, Zookeeper, TCP/IP, REST APIs, Django, Flask

Languages: English (Native) & French (Native)

Interests & Hobbies: Dance, Piano, Figure Skating, DragonBoat, Basketball, Volleyball, Drawing, Reading, Travelling